

Session 1

The Balance of Payments and External Sustainability

A euro-area reading of Schmitt-Grohe, Uribe and Woodford, chapters 1 and 2

International Macroeconomics of the Euro Area | 3 hours | CY Cergy Paris Université

How the three hours are built

Four blocks, three exercises, two long breaks

●	0:00	Opening questions	10 min
●	0:10	Block 1 What the accounts measure	35 min
●	0:45	Break	10 min
●	0:55	Block 2 Exercise A and debrief	40 min
●	1:35	Break	10 min
●	1:45	Block 3 From flow to stock, Exercise B	25 min
●	2:10	Stretch	5 min
●	2:15	Block 4 Can this go on, Exercise C	40 min
●	2:55	Wrap and problem set 1	5 min

Two questions before we start

Hands up. No data, no phones.

1

Which EU country runs the largest current account surplus measured in euros?

2

Which runs the largest surplus measured against the size of its own economy?

If the two answers are different countries, why would that be?

The answers split

Current account balances, 2025

Germany

EUR 202.7 bn

the largest surplus
in absolute terms

Denmark

12.5% of GDP

the largest surplus
relative to the economy

Romania

-7.9% of GDP

the largest deficit
relative to the economy

- Malta (8.6%) and Ireland (8.2%) follow Denmark; Cyprus (-6.4%) and Greece (-5.7%) follow Romania.
- Sixteen EU countries ran a surplus in 2025 and eleven ran a deficit. In 2022 nineteen ran a deficit.
- So the question is not who is in surplus today. It is what a surplus or a deficit actually is.

1

What the accounts measure

Chapter 1, slides 6 to 19, re-anchored on Europe

35 minutes

The balance of payments has three accounts

A complete record of transactions between residents and non-residents

1

Current account

Goods, services, income and transfers.
What the country earns and spends abroad this year.

This is where we spend today

2

Financial account

Transactions in assets and liabilities.
How the current account is financed.

Session 4, and TARGET2

3

Capital account

Acquisition of non-produced assets,
capital transfers, debt forgiveness.

Small, but not zero in the EU

The current and capital accounts together tell you an economy's exposure to the rest of the world. The financial account tells you how it is paid for.

The current account is a sum of four things

Everything in this session hangs off this one line

$$\text{CA} = \text{Goods} + \text{Services} + \text{Primary income} + \text{Secondary income}$$

Goods

Exports minus imports of merchandise

*Germany, Ireland,
Netherlands in surplus*

Services

Transport, travel, finance, business services

*Spain, Luxembourg,
Malta in surplus*

Primary income

Income earned on foreign assets, minus income paid on foreign liabilities

*Depends on who owns
what, not on trade*

Secondary income

Transfers with nothing received in return: EU funds, remittances, aid

*Large and positive for
several newer members*

Trade balance = goods + services. Note that the trade balance is only part of the current account.

Primary income: the account that trade cannot explain

Income on claims, not on goods

What it records

- Interest, dividends and reinvested profits received from abroad
- Minus the same paid out to non-residents
- Plus wages of cross-border workers

Why it matters in Europe

- A country with a large stock of foreign assets earns on it
- A country hosting foreign multinationals pays out on it
- Same trade flows, opposite income balances

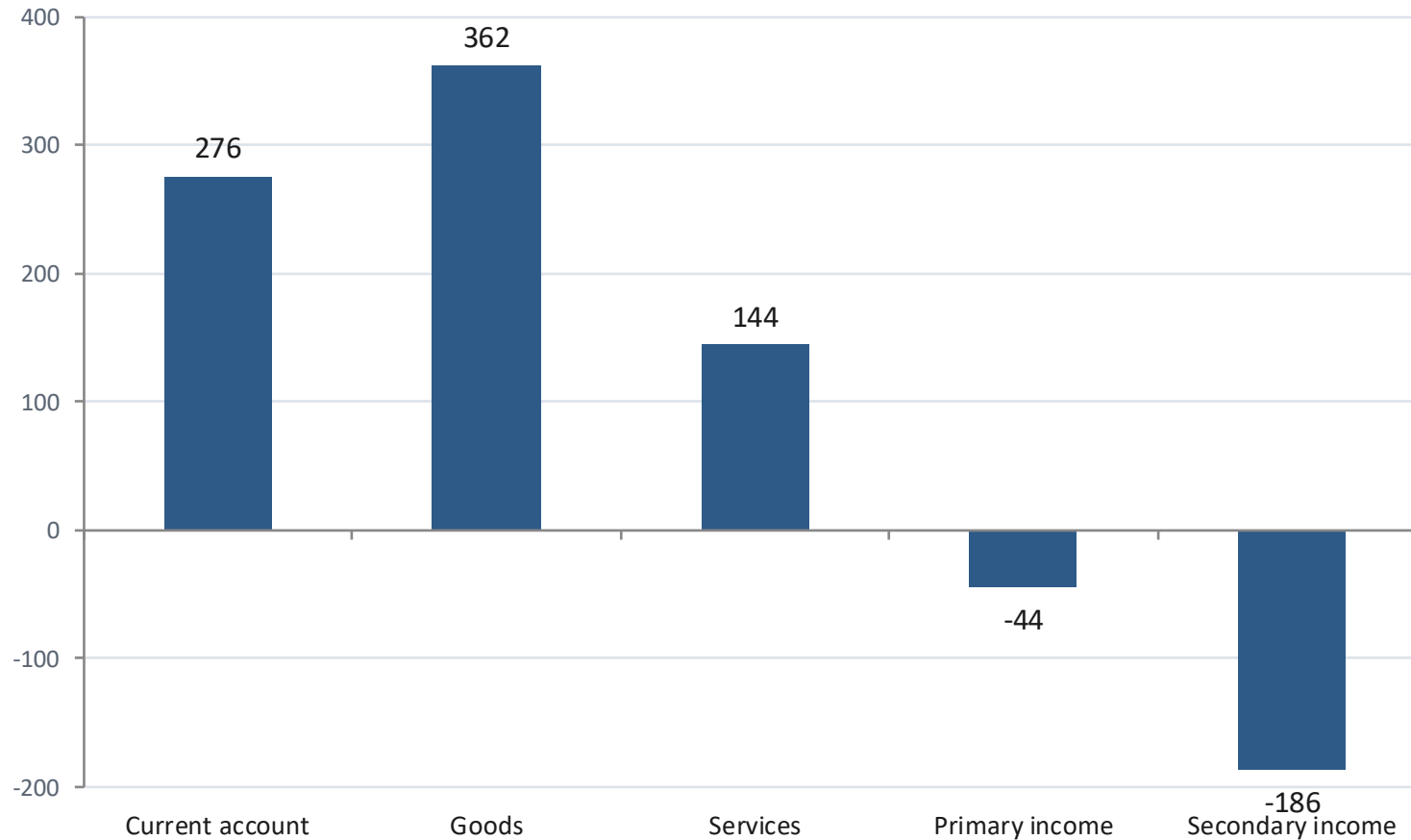
Two European faces of the same account

Germany owns a very large stock of foreign assets, so its income balance adds to an already large goods surplus.

Ireland hosts the foreign parent companies' profits, so its income balance subtracts heavily from an enormous goods surplus.

The euro area in 2025

Current account with the rest of the world, EUR billion

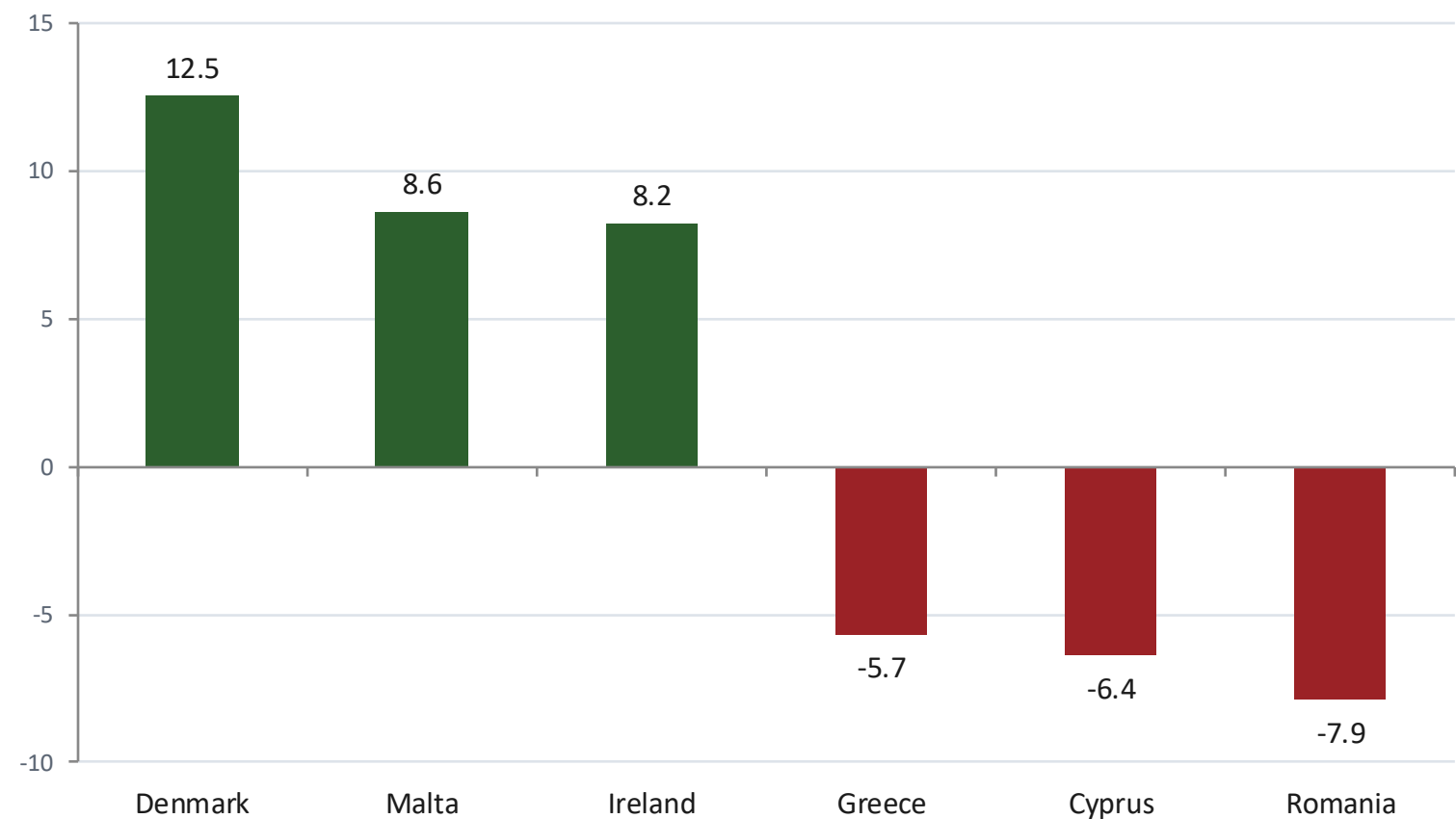


Read it off the identity

- A large goods surplus and a solid services surplus
- Both income accounts are negative
- The two offsets are big enough to cut the surplus by about two fifths
- The bloc as a whole is a net lender to the rest of the world

Same rules, very different external positions

Current account balance, 2025, percent of GDP. EU member states.



The question for today

A spread of twenty points of GDP inside one single market, and for most of these countries inside one currency.

Is the deficit end of this chart a problem?

Chapter 2 gives the answer, and it is not the one most people expect.

Source: Eurostat, bop_gdp6_q. Extremes of the 2025 distribution. Denmark and Romania are EU members outside the euro; Malta, Ireland, Greece and Cyprus are inside it.

Three things to carry into the exercise

Block 1 in one slide

1

The trade balance is not the current account

Goods and services are two of four components. The other two are income accounts and they are not small.

2

A surplus is not automatically good

It is a statement that the country is lending to the rest of the world this year. Whether that is wise depends on why.

3

Everything must add up

The four components sum to the current account exactly. If your numbers do not, your data selection is wrong, not the identity.

Break, 10 minutes

When we come back, open ec.europa.eu/eurostat in a browser.
You will need it for the next forty-five minutes.

2

Exercise A: your own current account

You build the table that the textbook prints for the United States

40 minutes, 25 working and 15 discussing

Exercise A: the recipe

Eurostat Data Browser, dataset bop_gdp6_q

1

Open the dataset

Search the Eurostat Data Browser for bop_gdp6_q, "Main Balance of Payments and International Investment Position items as share of GDP".

2

Set the filters

Frequency: annual. Partner: rest of the world. Stock or flow: balance. Unit: percentage of GDP.

3

Choose your country

Your own, whichever passport or home university you came here with. If nobody in the room is Irish or Greek, two volunteers take those.

4

Select five items

Current account, goods, services, primary income, secondary income. Read off the most recent year.

Ten minutes. Work in pairs, and pair up with someone from a different country.

Fill this in, then check it

Your country, latest available year

Current account

% of GDP

Goods

% of GDP

Services

% of GDP

Primary income

% of GDP

Secondary income

% of GDP

The check

**Goods + Services
+ Primary + Secondary**

= Current account

If it does not, go back to the filters. The identity holds by construction, in every country, in every year.

Extension for the algebra track: download the CSV, add 2008 next to the latest year, and compute the change in each component. For Spain and Portugal the swing from deep deficit to surplus is the single most important fact in this course.

What the board should look like

Six countries, six different reasons for the same-looking number

	Goods	Services	Primary	Secondary	What drives it
Germany	+	-	+	-	Goods surplus, reinforced by income on a huge stock of foreign assets
Ireland	++	-	--	-	Vast goods surplus, most of the profit paid out to foreign parents
Greece	-	+	-	+	Goods deficit, offset by shipping and tourism services
Poland	+	+	-	+	EU transfers and remittances show up in secondary income
Luxembourg	-	++	--	-	Almost entirely a financial services economy
Spain	-	++	-	-	Goods deficit more than covered by tourism and services

Signs are indicative of the recent pattern. Students replace them with the year they pulled.

The fact: Ireland, 12 July 2016

Ninety seconds, then straight back to the accounts

1 12 July 2016, 11am

The CSO restates 2015

Real GDP growth for 2015 is revised from about 7.8 percent, estimated in March on incomplete data, to 26.3 percent.

2 Within a day

The number makes world news

Commentators abroad mock the figure. Irish economists object: the CSO had followed the international standard exactly.

3 What actually happened

Balance sheets relocated

The CSO attributes the shift to relocations of entire corporate balance sheets, intellectual property included, into Ireland.

4 July 2017

Ireland invents GNI*

A second measure strips out the globalisation effects. On GNI*, real growth in 2015 was minus 0.4 percent.

Three questions, one answer

So what did it measure? **GDP +25.2%** **GNI +13.7%** **GNI* -0.4%**

Did Ireland produce 26% more goods?

No

Did Irish incomes rise 26%?

No

Was Ireland 26% more productive?

No

Ireland: what is a current account actually measuring?

The case that breaks the trade intuition

Headline current account

EUR 91 bn

As published in the balance of payments, 2024

Modified current account, CA*

EUR 25 bn

The Irish statistical office's own adjusted measure

- Ireland's measured goods surplus was about 36 percent of GDP in 2025, by far the largest in the EU.
- Foreign multinationals book profits in Ireland, and those profits flow back out through primary income.
- The CSO publishes CA* precisely because the headline number says little about Irish residents' income.
- The lesson is general: a balance of payments number is a measurement, and measurements have conventions.

Three reasons the trade balance and the current account come apart

And each one has a European poster child



A large stock of foreign assets

Income received exceeds income paid, so the current account sits above the trade balance.

Germany, Netherlands



A large stock of foreign liabilities

Income paid exceeds income received, so the current account sits below the trade balance.

Ireland, Greece, Hungary



Large net transfers received

EU structural funds and worker remittances add to the current account with no trade flow at all.

Poland, Romania, Bulgaria

Notice that two of the three are about stocks, not flows. That is the bridge into the next block.

Break, 10 minutes

Keep Eurostat open. There is a short data exercise in the next block too.



3

From flow to stock

Chapter 1, slides 28 to 59, without the US framing

25 minutes

The current account is a flow, the position is a stock

Net international investment position: NIIP = foreign assets minus foreign liabilities

The water tank



The level in the tank is the position. The pipes are the annual current account.

Where Europe stands

Greece	about -133%
Portugal	about -51%
Germany	strongly positive

NIIP as a share of GDP, early 2026. Greece owes the rest of the world well over a year of national output.

But the tank leaks

The current account is not the only thing that moves the position

$$\text{Change in NIIP} = \text{Current account} + \text{Valuation changes}$$

What moves valuations

- Exchange rate moves, when assets and liabilities are in different currencies
- Equity prices, at home and abroad
- Bond prices, so interest rate moves
- None of these is a transaction. Nothing crossed a border.

A euro-area example

A German fund holds USD 100 of US equities. Its liabilities are EUR 90 of euro-denominated bonds held by non-residents.

The euro falls from 1.20 to parity with the dollar.

The asset is now worth more in euros. The liability is unchanged. The NIIP improves, and not one euro of trade took place.

Exercise B: does the tank behave?

Ten minutes, same country as Exercise A

1

Open dataset tipsii10, net international investment position, annual, percent of GDP.

2

Chart your country from 2000 to the latest year.

3

Put it next to the current account series you already pulled in Exercise A.

4

Answer this: in every year the current account was negative, did the position get worse?

The answer is no. The residual between the two series is the valuation change. You have just reproduced the hypothetical NIIP figure from chapter 1, for a country you care about.

Extension

Pull the current account in euro millions rather than percent. Cumulate it from 2000 and add it to the year-2000 position.

Compare with the actual position. The gap is the cumulative valuation effect.

In Stata this is a sort, then `gen cum = sum(ca)`.

A puzzle worth ending the block on

Position and income do not always have the same sign

The textbook version

The United States is the world's largest net debtor, and yet it receives more investment income than it pays.

Two explanations: unrecorded intangible assets, or a return differential on risky assets against safe liabilities.

The European mirror

Germany holds an enormous positive position and earns a strikingly modest return on it.

The question is symmetric: what a country owns matters, but so does what kind of asset it owns.

Why this matters for the rest of the course

If the return on what you own differs from the return on what you owe, then the cost of carrying an external position depends on its composition, not only its size. Hold that thought until session 4, when euro sovereign spreads reopen the same question inside the union.

Stretch, 5 minutes

Last block. We finally get to ask whether any of this is sustainable.

4

Can this go on?

Chapter 2 in full, model and all

40 minutes

Can a country run a trade deficit forever?

Chapter 2 answers this with four lines of algebra, so we will do the algebra

If it is a net creditor

Yes. The interest earned on what it already owns abroad pays for the imports. The deficit can run forever.

If it is a net debtor

No. At some point it has to run a trade surplus, because the debt has to be serviced out of real resources.

Everything hangs on the sign of the starting position

The next three slides build the argument. It is ordinary substitution, no calculus, and both tracks should follow it. Only the extension to a growing economy at the end is set aside for the starred problem-set question.

Step 1: set the model up

Two periods, a starting position B_0 , and an interest rate r

Assumed throughout: the income balance is just net investment income, net unilateral transfers are zero, valuation changes are zero.

Net investment income in period 1 = $r B_0$

What the starting position earns, or costs, in period 1

$B_1 = (1 + r) B_0 + TB_1$

The position at the end of period 1

(1)

$B_2 = (1 + r) B_1 + TB_2$

The same accumulation, one period later

(2)

$B_2 = 0$

Nobody lends into period 3, so nothing can be left outstanding

(3)

Step 2: substitute (1) into (2), then impose (3)

One line of algebra, and the answer falls out

$$(1 + r) B_0 = -TB_1 - TB_2 / (1 + r) \quad (4)$$

In words: the starting position, including the interest on it, equals the present discounted value of the trade deficits the country can afford to run.

$B_0 < 0$

Net debtor

The right-hand side must be negative, so at least one TB has to be positive. A surplus is unavoidable, though not necessarily now.

$B_0 > 0$

Net creditor

The right-hand side can be positive throughout. Trade deficits in both periods are affordable, financed by interest income.

The same result holds for any number of periods, including an infinite number.

Step 3: ask the same question about the current account

The current account is the change in the position, so the argument repeats

$$CA1 = B1 - B0$$

Period 1: the current account is the change in the position

$$CA2 = B2 - B1$$

Period 2: the same again

$$B2 = 0$$

The terminal condition from step 1

Eliminate B1

$$B0 = -CA1 - CA2$$

A country can run current account deficits in both periods only if it started out as a creditor. True for any finite number of periods.

Step 4: where $CA = S - I$ comes from

Five substitutions in the national accounts, nothing more

1 $Q + IM = C + G + I + X$

Supply equals demand: output plus imports, against the four uses

2 Add $r B(t-1)$ to both sides, and use $TB = X - IM$

Rearranging the trade terms into a single balance

3 $Y = Q + r B(t-1)$ and $CA = r B(t-1) + TB$

National income, and the current account as income plus trade

4 $Y = C + G + I + CA$

Substituting both definitions back in

5 $S = Y - C - G$ and $A = C + G + I$

National saving, and domestic absorption

$$CA = S - I$$

The gap between saving and investment

$$CA = Y - A$$

The gap between income and absorption

The same current account, written four ways

All four are identities. None of them is a theory.

$$CA = TB + r B$$

The trade balance plus income on the existing position

$$CA = B(t) - B(t-1)$$

The change in the net foreign asset position

$$CA = S - I$$

The gap between national saving and domestic investment

$$CA = Y - A$$

The gap between national income and domestic absorption

The third one is the one that will do the most work later: a deficit is a country investing more than it saves.

A growing economy changes the answer

You do not have to repay. You have to stop the ratio rising.

$$\text{Trade balance / GDP} = - [(r - g) / (1 + g)] \times \text{NIIP / GDP}$$

If r is below g

A net debtor can run trade deficits forever and still see the debt ratio fall. Growth does the work.

If r equals g

A zero trade balance holds the ratio steady, whatever its level.

If r exceeds g

The required surplus is proportional to the debt. The bigger the position, the harsher the arithmetic.

This is the infinite-horizon case mentioned at the end of chapter 2. It is the version that matters for real countries, and it is the version we will use.

Exercise C: Greece's arithmetic

Fifteen minutes, pen and paper

Greece's position is about -133% of GDP.

What trade balance does it need to hold that ratio constant?

$r - g$	Required TB / GDP
0 %	
1 %	
2 %	
3 %	

Then discuss

- Greece ran a current account deficit of 5.7% of GDP in 2025 and a goods deficit of 13.6%.
- Compare that to the numbers you just computed. How far apart are they?
- Now the important question: what is r for Greece?
- Most of its external liabilities are official-sector loans at concessional rates, not market debt.

So: is the deficit end of that chart a problem?

Back to the question from block 1

**It depends on the interest-growth differential and on who holds the claim.
Not on the size of the deficit in any single year.**

Who holds it

Official creditors at concessional rates behave very differently from private bondholders who can run.

At what rate

A small change in r relative to g changes the required surplus by several points of GDP.

In whose currency

Inside a monetary union there is no exchange rate to adjust. That constraint arrives in session 3.

Before you go



Problem set 1

Questions 1 to 4 are graphical and required. Questions 5 and 6 are starred: the two-period budget constraint and the debt-stabilising trade balance. Extra credit, not required for full marks.



Reading

Schmitt-Grohe, Uribe and Woodford, chapters 1 and 2. Skim chapter 3 before the next session.



Keep your data

You will reuse the series from exercises A and B in session 2 and again in session 6.



Next session

Chapter 3: the intertemporal theory of the current account, and TARGET2 as the financial-account counterpart to what we saw today.